

Steadier, Not Closer: Separating Convergence from Crystallization in Deliberative Polls

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September 22, 2026

Abstract

Observed opinion dispersion can fall because true opinions converge or because answers become less noisy. We derive the decomposition for repeated multi-item attitude measures in partially nested and multiple-membership deliberative designs. Wave-varying measurement error contaminates within-group homogenization, the intraclass correlation, and regressions of change on prior opinion. Under classical independent error it cannot, however, manufacture an increase in the absolute variance of group means. Multiple invariant indicators, random assignment to groups, and a survey-only control arm identify distinct estimands for true convergence, between-group divergence, and crystallization.

1 Setup

Let $a \in \{0, 1\}$ index control and deliberation, $t \in \{1, 2\}$ wave, i person, g assigned group, and k item. The measurement model is

$$y_{kita} = \nu_k + \lambda_k \eta_{ita} + \epsilon_{kita}, \quad \mathbb{E}(\epsilon_{kita} \mid \eta_{ita}, a, g) = 0. \quad (1)$$

Loadings and intercepts are invariant across waves and arms; residual variances θ_{kta} are free. For fixed composite weights q normalized so that $\sum_k q_k \lambda_k = 1$, define

$$S_{ita} = \sum_k q_k (y_{kita} - \nu_k) = \eta_{ita} + u_{ita}, \quad \omega_{ta} = \text{Var}(u_{ita} \mid a). \quad (2)$$

The scalar $\omega_{ta} = q' \Theta_{ta} q$ includes residual covariance when it is modeled. “Crystallization” is a treatment-caused reduction in ω .

2 What changing reliability does and does not bias

Proposition 1 (The mean effect survives heteroskedastic measurement error). *If measurement errors have conditional mean zero, then*

$$\mathbb{E}[(S_{i2} - S_{i1}) \mid a = 1] - \mathbb{E}[(S_{i2} - S_{i1}) \mid a = 0] = \mathbb{E}[(\eta_{i2} - \eta_{i1}) \mid a = 1] - \mathbb{E}[(\eta_{i2} - \eta_{i1}) \mid a = 0].$$

Equality holds without requiring $\omega_{1a} = \omega_{2a}$.

Proof. Substitute $S_{it} = \eta_{it} + u_{it}$ and apply iterated expectations. Each error mean is zero, so both arm-specific mean error changes vanish. Different error variances change the sampling variance, not the estimand. $\square$

Proposition 2 (Observed within-group homogenization mixes two mechanisms). *Under local independence of u across people and independence from true opinion, the population within-group variance satisfies*

$$W_{tga}(S) = W_{tga}(\eta) + \omega_{ta}.$$

Consequently,

$$W_{2g1}(S) - W_{1g1}(S) = \{W_{2g1}(\eta) - W_{1g1}(\eta)\} + (\omega_{21} - \omega_{11}).$$

Proof. Within a group, $\text{Var}(\eta + u) = \text{Var}(\eta) + \text{Var}(u) + 2\text{Cov}(\eta, u)$. The covariance is zero by assumption. Subtract the two wave-specific identities. $\square$

Corollary 1. *If true within-group dispersion is unchanged and $\omega_{21} < \omega_{11}$, observed variance falls. The ratio statistic $H = 1 - W_2(S)/W_1(S)$ is positive even though true convergence is zero.*

Proposition 3 (Classical crystallization cannot create absolute divergence). *For balanced groups of size n , independent person-level error implies*

$$\text{Var}_g(\bar{S}_{gt}) = \text{Var}_g(\bar{\eta}_{gt}) + \frac{\omega_t}{n}.$$

Holding true group means fixed, a fall in ω_t weakly lowers the absolute variance of observed group means. With unequal sizes the error contribution for group g is ω_t/n_g .

Proof. $\bar{S}_{gt} = \bar{\eta}_{gt} + n^{-1} \sum_i u_{igt}$. Independence makes the cross covariance zero and the variance of the averaged error ω_t/n . $\square$

Remark 1. *This proposition is a useful boundary. Crystallization can create apparent within-group homogenization, but not an increase in the absolute variance of group means under the stated assumptions. A claimed absolute divergence artifact therefore requires nonclassical correlated error, changing group sizes or composition, scale noninvariance, or a statistic other than absolute variance.*

Proposition 4 (Relative divergence can rise mechanically). *If $\eta_{igt} = \mu_t + b_{gt} + v_{igt}$ with variances σ_{bt}^2 and σ_{wt}^2 , then*

$$\text{ICC}_t(S) = \frac{\sigma_{bt}^2}{\sigma_{bt}^2 + \sigma_{wt}^2 + \omega_t}.$$

For $\sigma_{bt}^2 > 0$, its derivative with respect to ω_t is negative. Thus reducing error raises the observed ICC even when both true variance components are fixed.

Proof. The group covariance of two distinct people is σ_b^2 , while each observed variance adds ω_t to the true total. Differentiate the ratio. $\square$

Proposition 5 (Observed prior opinion induces regression to the mean). *Under classical error independent across waves,*

$$\text{Cov}(S_{i2} - S_{i1}, S_{i1}) = \text{Cov}(\eta_{i2} - \eta_{i1}, \eta_{i1}) - \omega_1.$$

Proof. Expand the covariance of $(\Delta\eta_i + u_{i2} - u_{i1})$ and $(\eta_{i1} + u_{i1})$. All cross terms vanish except $\text{Cov}(-u_{i1}, u_{i1}) = -\omega_1$. $\square$

3 Structural decomposition

For a fixed group, write $x_i = \eta_{i1} - \bar{\eta}_{g1}$. The treated change model is

$$\eta_{i2} = \eta_{i1} + \tau + \gamma(\bar{\eta}_{g1} - \eta_{i1}) + b_g + \delta_i.$$

After subtracting the T2 group mean,

$$\eta_{i2} - \bar{\eta}_{g2} = (1 - \gamma)x_i + (\delta_i - \bar{\delta}_g).$$

When x_i and δ_i are independent, the inherited within-group variance is multiplied by $(1 - \gamma)^2$; idiosyncratic true change may offset that reduction. The common effect b_g cancels within group and contributes to between-group variance. Hence γ , σ_b^2 , and $\omega_2 - \omega_1$ correspond to different mechanisms.

With reshuffling, let W contain session weights and b the session effects. Then

$$\text{Var}(Wb) = \sigma_b^2 WW',$$

so two participants' group-induced changes covary in proportion to the overlap of their session weights. One-hot rows recover fixed groups.

4 Identification

With at least three locally independent indicators and a scale constraint, off-diagonal item covariances identify the common factor variance because $\text{Cov}(y_k, y_\ell) = \lambda_k \lambda_\ell \text{Var}(\eta)$ for $k \neq \ell$. Indicator variances then identify residual variances. Equality of loadings and intercepts across waves makes the latent scale comparable; equality of residual variances is not imposed and is precisely the restriction being studied [3]. Same-item residual correlations across waves must be modeled or tested rather than silently set to zero.

Random assignment to groups makes baseline latent composition independent of group effects. A survey-only control arm identifies the error-variance change from retesting and panel conditioning. The crystallization estimand is

$$C = (\omega_{21} - \omega_{11}) - (\omega_{20} - \omega_{10}).$$

Interpreting C causally additionally requires measurement invariance, no differential attrition, and no treatment-induced reporting process omitted from the measurement model. Single-item outcomes observed twice do not identify both latent change and two error variances without further information.

The design is partially nested because treatment participants share deliberation groups while controls do not [1, 4]. For multiple sessions, the multiple-membership extension has covariance

$$\sigma_b^2 WW'.$$

5 Simulation check

The accompanying simulation fits conventional random-intercept models with `lme4`; it does not substitute a hand-written variance-component estimator. It varies convergence, between-group divergence, and item-error reduction independently. Table 1 reports 200 replicates per scenario. In the crystallization-only condition, observed within-group spread falls and the observed ICC rises while true spread and true between-group variance remain stable. Raw group-mean variance falls, as the third proposition requires.

Table 1: Simulation decomposition. Entries are means and 95% simulation intervals.

Scenario	Quantity	Estimate
Crystallization only	Observed within-group homogenization	0.139 [0.083, 0.196]
Crystallization only	True within-group homogenization	0.000 [0.000, 0.000]
Crystallization only	Observed between-component change	-0.001 [-0.030, 0.025]
Crystallization only	True between-component change	0.000 [0.000, 0.000]
Crystallization only	Observed ICC change	0.013 [-0.007, 0.032]
Convergence only	Observed within-group homogenization	0.463 [0.423, 0.499]
Convergence only	True within-group homogenization	0.510 [0.510, 0.510]
Convergence only	Observed between-component change	-0.000 [-0.031, 0.028]
Convergence only	True between-component change	0.000 [-0.021, 0.016]
Convergence only	Observed ICC change	0.071 [0.039, 0.106]
Divergence only	Observed within-group homogenization	-0.002 [-0.065, 0.056]
Divergence only	True within-group homogenization	-0.000 [-0.000, 0.000]
Divergence only	Observed between-component change	0.233 [0.138, 0.356]
Divergence only	True between-component change	0.233 [0.134, 0.357]
Divergence only	Observed ICC change	0.141 [0.099, 0.185]
All three	Observed within-group homogenization	0.564 [0.535, 0.592]
All three	True within-group homogenization	0.510 [0.510, 0.510]
All three	Observed between-component change	0.233 [0.138, 0.351]
All three	True between-component change	0.234 [0.133, 0.353]
All three	Observed ICC change	0.307 [0.234, 0.373]

6 Implications

Variance-based claims in deliberative polling should distinguish three targets: true within-group convergence, absolute between-group divergence, and reduced measurement error. The first and third are inseparable from a single noisy index. The second is not mechanically inflated by classical error reduction, although an ICC or between/within ratio is. This distinction narrows the model's burden and clarifies what a reanalysis of the twenty-one-poll archive can and cannot revise [2].

References

- [1] Daniel J. Bauer, Sonya K. Sterba, and Denise D. Hallfors. Evaluating group-based interventions when control participants are ungrouped. *Multivariate Behavioral Research*, 43(2):210–236, 2008. doi: 10.1080/00273170802034810.
- [2] Robert C. Luskin, Gaurav Sood, James S. Fishkin, and Kyu S. Hahn. Deliberative distortions? homogenization, polarization, and domination in small group discussions. *British Journal of Political Science*, 52(3):1205–1225, 2022. doi: 10.1017/S0007123421000168.
- [3] John J. McArdle. Latent variable modeling of differences and changes with longitudinal data. *Annual Review of Psychology*, 60:577–605, 2009. doi: 10.1146/annurev.psych.60.110707.163612.
- [4] Chris Roberts and Rebecca Walwyn. Design and analysis of non-pharmacological treatment trials with multiple therapists per patient. *Statistics in Medicine*, 32(1):81–98, 2013. doi: 10.1002/sim.5520.